

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – II

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Class: TYIT	Batch: 1
Date of Assignment: 31-07-2026	Date/Time of Submission: 31-07-2026

Working with Data Sources and Data Preparation in Tableau ..

a) Connect to different data sources (Excel, CSV, Database)

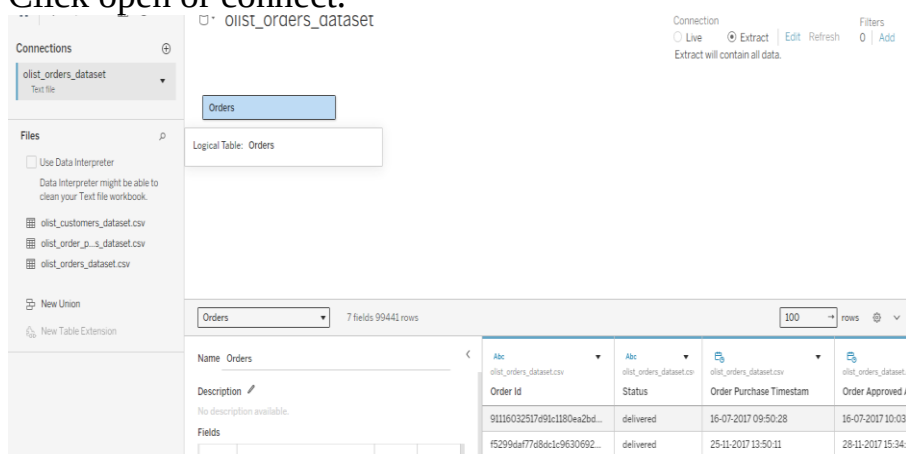
Step 1:

Open Tableau Desktop.

Click **Connect** or Text file

Select **Microsoft Excel**, **Text File (CSV)**, or **Database**.

Click open or connect.



b) Create and use data extracts vs live connections

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'olist_orders_dataset' as a Text file. Below it, the 'Files' pane shows a list of CSV files: 'olist_customers_dataset.csv', 'olist_order_p_s_dataset.csv', and 'olist_orders_dataset.csv'. The 'Orders' logical table is selected. The main view displays a table with 7 fields and 99441 rows. The fields are: Order Id, Status, Order Purchase Timestamp, and Order Approved At. The data shows orders with status 'delivered' and purchase timestamps from 2017.

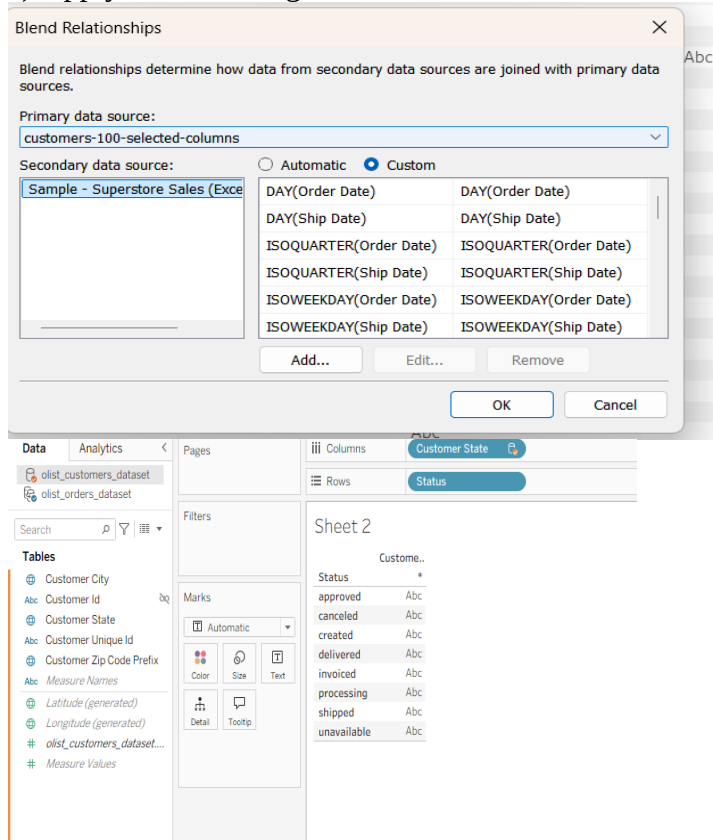
c) Manage metadata (rename, hide, change data types)

The screenshot shows the 'Change Data Type' context menu for the 'Order Id' field. The menu options include: Aliases..., Create, Transform, Convert to Measure, Change Data Type (selected), Default Properties, Geographic Role, Image Role, Group by, Folders, Hierarchy, Replace References..., and Describe... The 'Change Data Type' sub-menu is open, showing options: Number (decimal), Number (whole), Date & Time (selected), Date, String, Spatial, Boolean, and Default. The 'Add to Sheet' sub-menu is also open, showing options: Duplicate, Rename, Hide (selected), Create, Convert to Continuous, Change Data Type, Default Properties, Group by, Folders, Hierarchy, Replace References..., and Describe... The background shows a table with columns like 'Order Id', 'Status', 'Order Purchase Timestamp', and 'Order Approved At'.

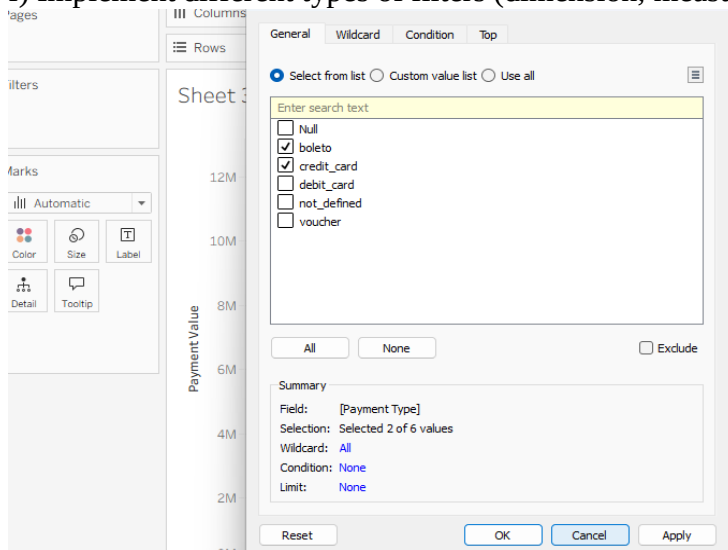
d) Perform joins (inner, left, right) on multiple tables .

The screenshot shows the Tableau Desktop interface with a join operation between 'olist_orders_dataset.csv' and 'olist_order_payments_dataset.csv'. The 'Join' dialog box is open, showing the 'Inner' join type selected. The 'Data Source' is 'olist_order_payments_dataset.csv'. The 'Join' button is highlighted. The main view displays a table with 13 fields and 10,388 rows. The fields are: Order Id, Customer Id, Status, Order Purchase Timestamp, and Order Approved At. The data shows orders with status 'delivered' and purchase timestamps from 2017.

e) Apply data blending between two sources .



f) Implement different types of filters (dimension, measure, date) .



g) Optimize performance using extracts and filtering..

Filter [Payment Value] ✕

 Range of values  **At least**  At most  Special

At least

0.01 13664.08

Show: Only Relevant Values ☐ Include Null Values

Reset OK Cancel Apply

Filter [Order Purchase Timestamp] ✕

 **Relative dates**  Range of dates  Starting date  Ending date  Special

Relative dates 31-07-2026 to 31-07-2026

Years Quarters Months Weeks **Days** Hours Minutes

☐ Yesterday ☐ Last days

☒ Today ☐ Next days

☐ Tomorrow ☐ Day to date

☐ Anchor relative to ☐ Include null values

Reset OK Cancel Apply

Filter [Status] X

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

- ☐ approved
- ☐ canceled
- ☐ created
- ☒ delivered
- ☐ invoiced
- ☐ processing
- ☐ shipped
- ☐ unavailable

All None ☐ Exclude

Summary

Field: [Status]
Selection: Selected 1 of 8 values
Wildcard: All
Condition: None
Limit: None

Reset OK Cancel

Edit Data Source Filters X

Filter	Scope	Details
Status	Orders and related tables	keeps delivered

Add... Edit... Remove

OK Cancel